

EC-390: Digital Signal Processing

LECTURE NO 5: *Mathematical Operations on Discrete Time Signals*

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Organization of Lecture

1. Time Shifting Operations.
2. Time Scaling Operations.
3. Amplitude Scaling Operations.

1. Time Shifting Operations

① Time Shifting

(i) Time Delay and Time Advance

- A signal $x(n]$ may be shifted in time by replacing the independent variable ' n ' by ' $n-m$ ' where ' m ' is an integer.

$n-m \Rightarrow$ Shifted signal

- If ' m ' is a positive integer, the time shift results in a delay by ' m ' units of time.

$n-(+m) = n-m =$ Delayed by ' m ' units

- If ' m ' is a negative integer, the time shift results in an advance by ' $|m|$ ' units of time.

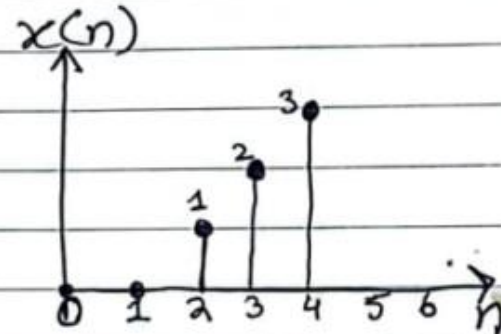
$n-(-m) = n+m =$ Advanced by ' $|m|$ ' units

1. Time Shifting Operations

Example 1:

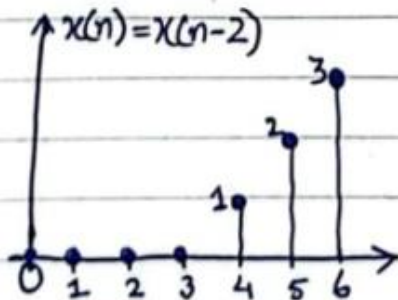
let $x(n) = 1$; $n=2$
 $=2$; $n=3$
 $=3$; $n=4$
 $=0$; for other n

Plot $x(n-2)$ and $x(n+2)$



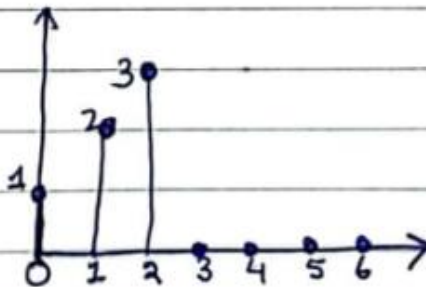
Soln:

Plotting $x(n-2)$:



(Delayed signal of $x(n]$)
(Move towards right by
'2' units)

Plotting $x(n+2)$:



(Advanced signal of $x(n]$)
(Move towards left by
'2' units)

1. Time Shifting Operations

(ii) Folding :

- Folding is also called as reflection.
- Folding means taking the mirror image of a signal. That means the signal is folded about time origin $n = 0$. Here independent variable ' n ' is replaced by ' $-n$ '.

1. Time Shifting Operations

Example 2:

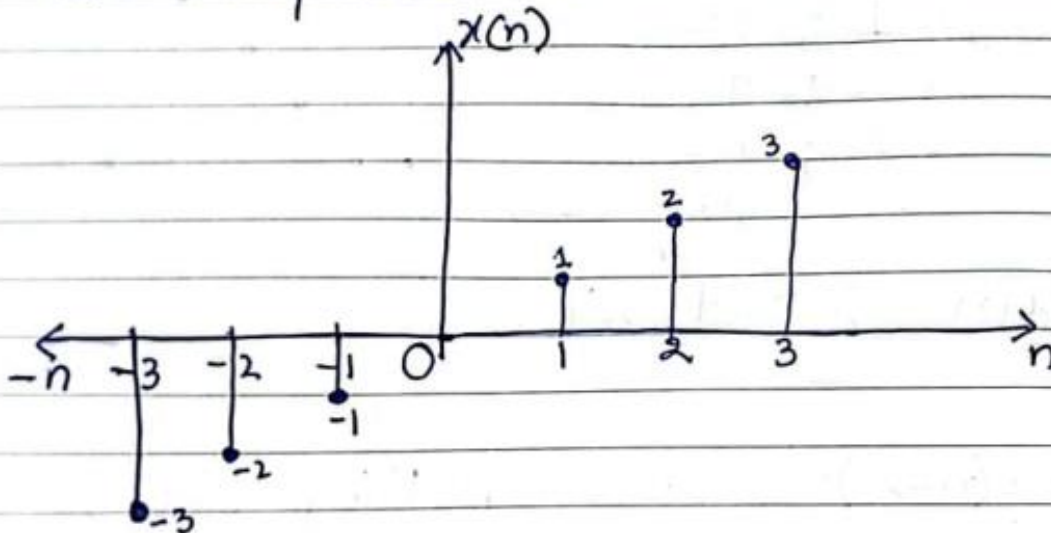
Let $x(n] = n; -3 \leq n \leq 3$, plot the folded signal $x(-n)$

Solⁿ:

The given signal $x(n)$ can be represented by a sequence as:

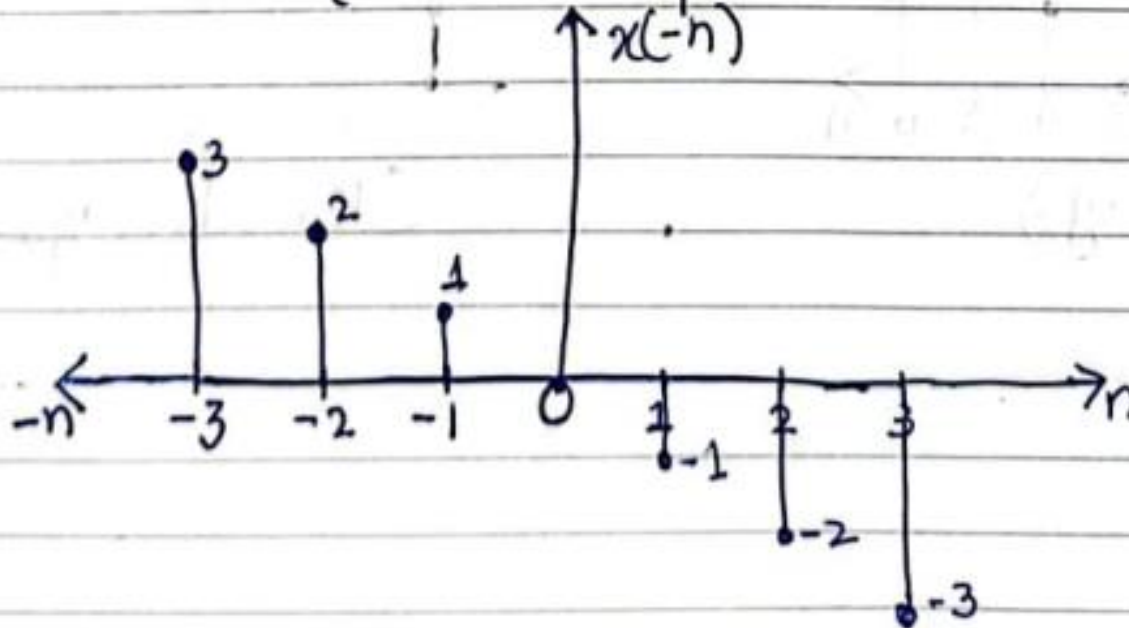
$$x(n) = \{-3, -2, -1, 0, 1, 2, 3\}$$

it can be plotted as:



1. Time Shifting Operations

The folded signal of $x(n]$ can be drawn as:

$$x(-n) = \{ 3, 2, 1, 0, -1, -2, -3 \}$$


1. Time Shifting Operations

(iii) Folding and Delay:-

$$x(n) \xrightarrow{\text{Folding}} x(-n) \xrightarrow{\text{Delayed by } m \text{ units}} x[-(n-m)] = x(-n+m)$$

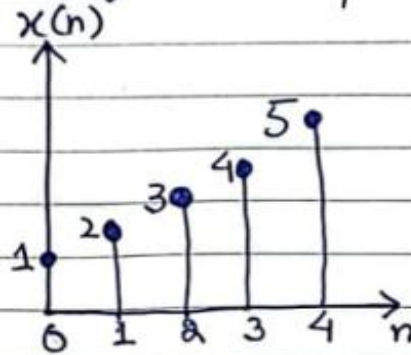
Example 3:-

$$\text{let } x(n) = \left\{ \underset{\uparrow}{1}, 2, 3, 4, 5 \right\} \text{ Plot } x(-n+2)$$

1. Time Shifting Operations

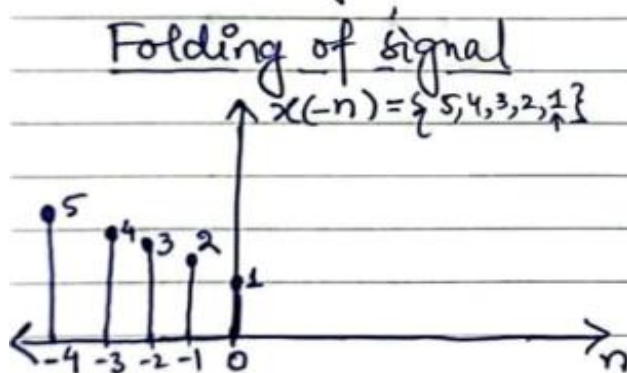
Solⁿ:

The signal $x(n]$ can be plotted as:

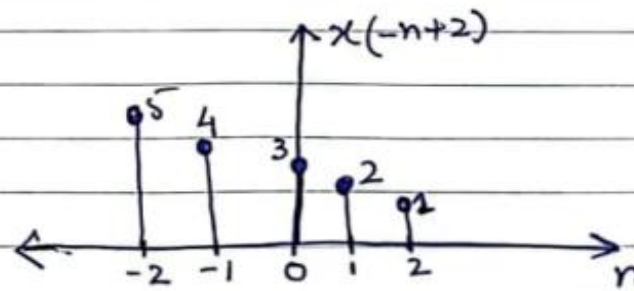


1st Perform the folding $x(-n)$ of a signal and then perform $x(-n+2)$

Folding of signal



Delaying folding signal by 2 units
or
Move towards right by '2' units



1. Time Shifting Operations

(iv) Folding and Advance:

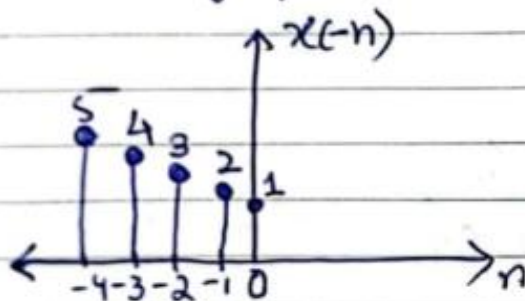
$$x(n) \xrightarrow{\text{Folding}} x(-n) \xrightarrow{\text{Advanced by 'm' units}} x[-(n+m)] = x(n-m)$$

Example 4: Same signal as in Example 3. Plot $x(-n-2)$

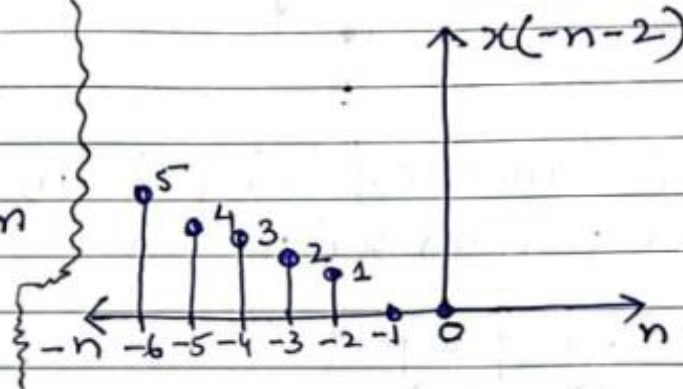
Solⁿ:

- 1st Performing the folding $x(-n)$
- Then perform $x(-n-2)$ i.e., advance the signal by '2' units.

Folding of $x(n)$



Advance folded signal by '2' units
or
Move towards left by '2' units



2. Time Scaling Operations

(2) Time Scaling:

- As the name indicates, time scaling operations are related to the change in time scale. There are two types of time scaling operations.
 - (i) Down scaling (Compression)
 - (ii) Upscaling (Expansion)

(i) Down Scaling:

- Consider the same sequence $x(n) = \{ \underset{\uparrow}{1}, 2, 3, 4, 5 \}$
 - Let $x(n)$ be an input to some device which produces output $y(n)$. Also say,
- $$y(n) = x(2n)$$
- Now from given sequence we can write,

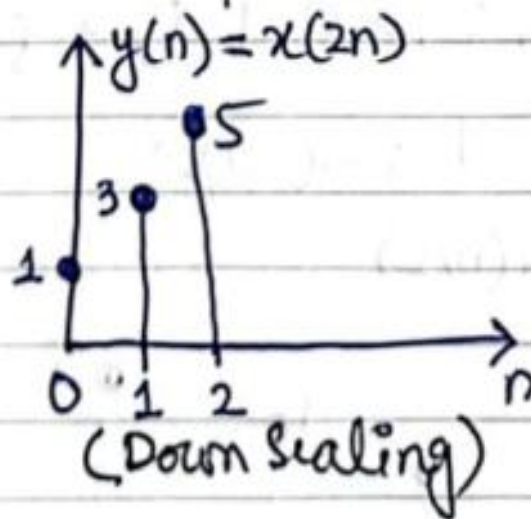
$$\begin{aligned} y(n) &= x(2n) \\ y(0) &= x(2(0)) = x(0) = 1 \\ y(1) &= x(2(1)) = x(2) = 3 \\ y(2) &= x(2(2)) = x(4) = 5 \\ y(3) &= x(2(3)) = x(6) = 0 \end{aligned}$$

2. Time Scaling Operations

- Therefore $y(n)$ can be represented as:

$$y(n) = \{ 1, 3, 5, 0, \dots \}$$

- $y(n)$ can be plotted as:



2. Time Scaling Operations

(ii) Up Scaling:

- consider the same sequence $x(n) = \{1, 2, 3, 4, 5\}$
- let $x(n)$ be an input to some device which produces output $y(n)$. Also say:
$$y(n) = x\left(\frac{n}{2}\right)$$

- Thus in this case:

$$y(0) = x\left(\frac{0}{2}\right) = x(0) = 1$$

$$y(1) = x\left(\frac{1}{2}\right) = x(0.5) = \text{No Sample} = 0$$

$$y(2) = x\left(\frac{2}{2}\right) = x(1) = 2$$

$$y(3) = x\left(\frac{3}{2}\right) = x(1.5) = \text{No Sample} = 0$$

$$y(4) = x\left(\frac{4}{2}\right) = x(2) = 3$$

$$y(5) = x\left(\frac{5}{2}\right) = x(2.5) = \text{No Sample} = 0$$

2. Time Scaling Operations

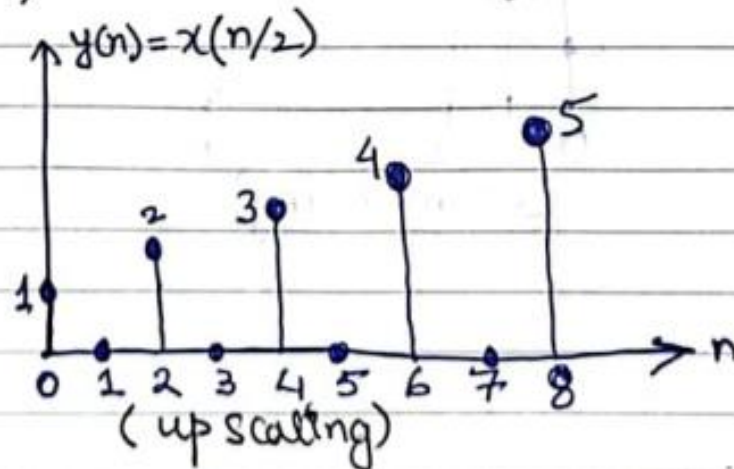
$$y(6) = x\left(\frac{6}{2}\right) = x(3) = 4$$

$$y(7) = x\left(\frac{7}{2}\right) = x(3.5) = \text{No Sample} = 0$$

$$y(8) = x\left(\frac{8}{2}\right) = x(4) = 5$$

- Thus $y(n) = \{1, 0, 2, 0, 3, 0, 4, 0, 5\}$

- Plot of $y(n)$ can be drawn as:



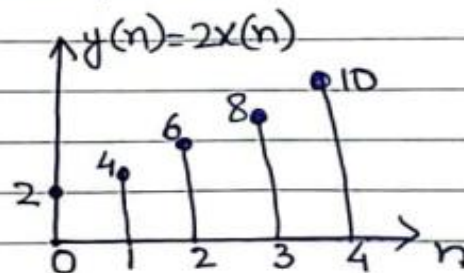
3. Amplitude Scaling Operations

③ Amplitude Scaling:

- As the name indicates, in case of amplitude scaling operations, amplitude of a signal is changed. Different amplitude operations are as follows:
 - i) Upscaling (Amplification)
 - ii) Downscaling (Attenuation)
 - iii) Addition
 - iv) Multiplication

(i) Upscaling:

- Consider same signal = $x(n) = \{ \underset{\uparrow}{1}, 2, 3, 4, 5 \}$
- Now let $y(n) = 2x(n)$
- This indicates multiplication of every amplitude by 2. That is:
$$y(n) = 2x(n) = \{ \underset{\uparrow}{2}, 4, 6, 8, 10 \}$$
- $y(n)$ is plotted as:



3. Amplitude Scaling Operations

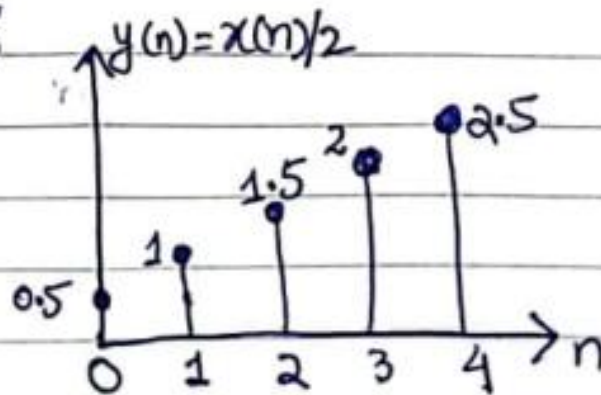
(ii) Downscaling:

- Consider the same signal $= x(n) = \{ \underset{\uparrow}{1}, 2, 3, 4, 5 \}$
- Now let $y(n) = \frac{1}{2} x(n)$ or $\frac{x(n)}{2}$

- This indicates the division of every amplitude by 2. that is :

$$y(n) = \frac{x(n)}{2} = \{ 0.5, 1, 1.5, 2, 2.5 \}$$

- Plot of $y(n)$ is :



3. Amplitude Scaling Operations

(iii) Addition:

- Consider two sequences as:

$$x_1(n) = \{1, 1, 0, 1, 1\}$$

$$x_2(n) = \{2, 2, 0, 2, 2\}$$

- let $y(n) = x_1(n) + x_2(n)$

$$y(n) = \{3, 3, 0, 3, 3\}$$

(iv) Multiplication:

- Consider two sequences (Same as in addition):

$$x_1(n) = \{1, 1, 0, 1, 1\}$$

$$x_2(n) = \{2, 2, 0, 2, 2\}$$

- let $y(n) = x_1(n) \times x_2(n)$

$$= \{2, 2, 0, 2, 2\}$$

References: Textbooks

1. *“Discrete-Time Signal Processing”, International Edition, by Oppenheim, Alan V; Schafer, Ronald W, (2013), Pearson.*
2. *“Digital Signal Processing Using Matlab”, 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related material from open source, mainly internet.*